

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Faculty of Technology and Engineering
Second Semester of M. Tech. (CE) Examination
May-2018

CE746 Cloud Computing

Date: 05.05.2018, Saturday Time: 01:30 p.m. To 04:30 p.m. Maximum Marks: 70

Instructions:

- (1) All questions are compulsory.
- (2) Indicate clearly, the option you attempt along with its respective question Number.
- (3) Figures to right indicate marks.
- (4) Rough work is done in the last page of main supplementary; don't write anything on the question paper

SECTION – I

Q - 1 Do as Directed.

- (a) Give the differences between Traditional computer and Virtual machine. **[03]**
- (b) Define the terms: **[04]**
 - (i) Heterogeneous Clouds
 - (ii) High-Throughput Computing (HTC)

Q - 2 Do as Directed.

- (a) **Answer the following questions. (Any Three)** **[12]**
 - 1. Show classification of Parallel and Distributed computing system with explanation.
 - 2. Explain the vCUDA for virtualization of general-purpose GPUs.
 - 3. What are the five abstraction levels in virtualization? Explain any two of them in detail.
 - 4. Explain following terms:
 - (i) Cloud Provider
 - (ii) Cloud Consumer
- (b) “A public cloud is owned by a third party and generally offers commercialized cloud services” True/False? Justify your answer. **[02]**

Q – 3

- (a) Draw and explain a layered diagram to relate the construction of IaaS, PaaS and SaaS clouds. **[07]**

- (b) Differentiate Physical and Virtual Clusters. [04]
- (c) “The system that results from the application of the Elastic Disk Provisioning pattern uses the thin- provisioning program to create virtual server disks” True/False? Justify your answer. [03]

OR

- (b) Define the steps of Live migration process of a VM from one host to another. [04]
- (c) How can redundant data within cloud storage devices be automatically avoided? [03]

SECTION – II

Q - 4 Do as Directed.

- (a) List cloud deployment models. Explain any one of them in detail. [04]
- (b) Show how common components of a data center working together to provide virtualized IT resources. [03]

Q - 5. Do as Directed.

- (a) **Answer the following questions. (Any Three)** [12]
1. How multitenant application can serve multiple cloud service consumers simultaneously?
 2. Explain Xen Architecture.
 3. As part of a failover system how to synchronizes failover system's data with the data in the primary cloud storage device?
 4. How can a virtual server survive the failure of its hosting hypervisor or physical server?

- (b) Define Apache CloudStack. [02]

Q - 6. Answer the following questions. (Any Two) [14]

1. Explain Direct I/O Access Architecture in detail.
2. How can a virtual server overcome performance limitations imposed by emulated file-based storage?
3. How can cloud providers securely store cloud consumer data on cloud storage devices?
4. Give detail description of Elastic resource capacity Architecture.